Daily Sequences Drill #7 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Mixed Synthesis Capstone · Every technique from Days 1–6, combined.

Section A Rapid Recognition

1. **Answer:** $5n - 1$

Method: $a = 4, d = 5$: $a_n = 4 + 5(n - 1) = 5n - 1$.

> **Investigate further:** Day 1's nth-term formula, unchanged all week.

2. **Answer:** 32

Method: $\sum_{k=0}^{5} \binom{5}{k} = 2^5 = 32$ (Day 2's row-sum identity, $x = 1$).

> **Investigate further:** Same trick as Day 2, A3.

3. **Answer:** 9

Method: $S_\infty = \dfrac{6}{1 - 1/3} = \dfrac{6}{2/3} = 9$ (Day 3).

> **Investigate further:** Same formula as Day 3, A5.

4. **Answer:** $x^2 - 7x + 10 = 0$

Method: Substitute $x^2, x, 1$ for a_n, a_{n-1}, a_{n-2} (Day 4).

> **Investigate further:** Roots: $(x - 2)(x - 5) = 0 \Rightarrow x = 2, 5$.

5. **Answer:** $x = 4$

Method: $x = 3x - 8 \Rightarrow -2x = -8 \Rightarrow x = 4$ (Day 5/Day 6's fixed-point technique).

> **Investigate further:** Same technique as Day 4 A7, Day 6 A3/A4.

6. **Answer:** Their sum must be a multiple of 5.

Method: Integer mean $\iff$ sum divisible by count (Day 6).

> **Investigate further:** Day 6's central definition, unchanged.

7. **Answer:** GP (ratio 3)

Method: $6/2 = 18/6 = 54/18 = 3$, constant ratio.

> **Investigate further:** Day 3's recognition skill.

8. Answer: Converges to 0.

Method: Both roots inside the unit circle $\Rightarrow a_n = Ar_1^n + Br_2^n \to 0$ (Day 4 D5, Day 5).

> **Investigate further:** The result both Day 4 and Day 5 converge on independently.

9. Answer: 10

Method: $\binom{5}{2} = 10$ (Day 2).

> **Investigate further:** Same computation as Day 2, A2/A5.

10. Answer: $x = \dfrac{c}{r-1}$

Method: Solve $x = rx - c$ for x (Day 6).

> **Investigate further:** The formula used throughout Day 6.

Section B Manipulation Drills

1. Answer: Verified: $a_n = 3n - 1$ satisfies $a_n = 2a_{n-1} - a_{n-2}$.

Method: $2a_{n-1} - a_{n-2} = 2(3(n-1)-1) - (3(n-2)-1) = 2(3n-4) - (3n-7) = 6n - 8 - 3n + 7 = 3n - 1 = a_n$. $\square$

> **Investigate further:** Every AP satisfies $a_n = 2a_{n-1} - a_{n-2}$ (characteristic equation $x^2 - 2x + 1 = (x-1)^2 = 0$, a repeated root at 1) — exactly Day 4's C7 result, now verified directly on a concrete AP.

2. Answer: A) $\frac{4}{3}$

Method: $(1-x)^{-1}$ at $x = \frac{1}{4}$: $\left(\frac{3}{4}\right)^{-1} = \frac{4}{3}$.

> **Investigate further:** Day 2's identity plus Day 3's substitution technique, combined exactly as in Day 3's own B10/C7.

3. Answer: A) Converges to 0.

Method: Both $\left(\frac{1}{2}\right)^n \to 0$ and $\left(\frac{1}{3}\right)^n \to 0$; with no forcing term, $a_n = A\left(\frac{1}{2}\right)^n + B\left(\frac{1}{3}\right)^n \to 0$.

> **Investigate further:** Day 4 D5's general fact, applied directly.

4. Answer: Fixed point $x = 1$ (repeated); yes, $a_1 = 1$ gives a constant sequence.

Method: $x = \dfrac{1}{2-x} \Rightarrow x(2-x) = 1 \Rightarrow 2x - x^2 = 1 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x-1)^2 = 0 \Rightarrow x = 1$ (unique, repeated). Since $a_1 = 1$ is exactly this fixed point, $a_2 = \dfrac{1}{2-1} = 1 = a_1$, and the sequence stays constant.

> **Investigate further:** A rational (Möbius-style) map with a REPEATED fixed point behaves differently from Day 5's period-3/period-4 examples, which had no repeated fixed points — worth noticing the distinction.

5. **Answer:** A) Summing a finite sequence.

Method: Both are instances of $\sum_k$(terms) — the underlying operation (summation) is identical, even though the term-generating rules (binomial coefficients vs. an arithmetic rule) are completely different.

> **Investigate further:** Recognising the SAME general operation underneath different-looking notation is a genuine synthesis skill, distinct from recognising when two things are actually unrelated (see B10).

6. **Answer:** A) A GP with ratio r.

Method: $b_n = a_n - \frac{c}{r-1}$ satisfies $b_{n+1} = rb_n$ exactly (Day 6's derivation, identical in form to Day 4's shift trick).

> **Investigate further:** This is literally Day 4's A5/A6 mechanism, applied to Day 6's alternating process.

7. **Answer:** $a_k = x \cdot a_{k-1}$, a first-order GP recurrence with (single) characteristic root x.

Method: Since $a_k = x^k$, clearly $a_k = x \cdot x^{k-1} = x \cdot a_{k-1}$.

> **Investigate further:** Every term of a geometric series is itself the closed-form solution of the simplest possible recurrence — Day 3 and Day 4 were secretly describing the same object from two different angles all along.

8. **Answer:** A) Every AP with $d \neq 0$ is monotonic.

Method: $a_{n+1} - a_n = d$ always, a fixed nonzero sign — increasing if $d > 0$, decreasing if $d < 0$, with no exceptions.

> **Investigate further:** Unlike Day 5's more exotic sequences (which needed real proof to establish monotonicity), an AP's monotonicity is immediate from its very definition.

9. **Answer:** A) No, unbounded.

Method: $|a_n| = |a||r|^n \to \infty$ when $|r| > 1$, regardless of the sign of a or r.

> **Investigate further:** This is the direct opposite of Day 3's convergence condition $|r| < 1$ — the two cases ($|r| < 1$ vs. $|r| > 1$) are exact mirror images.

10. **Answer:** Independent observations.

Method: Day 6's construction concerns DIVISIBILITY of a running (incrementally-built) sum by its own index; Day 2's row sum concerns the TOTAL of a fixed, complete row of binomial coefficients. Different objects, different questions, no structural connection beyond both being "a sum."

> **Investigate further:** Not every pair of similar-sounding facts from different days is actually related — correctly identifying when two ideas are genuinely independent, rather than forcing a false connection, is exactly as important as spotting real synthesis (contrast with B5).

Section C Substitution & Structure

1. **Answer:** A) They never coincide again.

Method: AP: $3, 7, 11, 15, 19, 23, \ldots$ GP: $3, 6, 12, 24, 48, \ldots$ Comparing term by term after the first: none match, and once the GP overtakes the AP (by term 4: $24 > 15$), the exponentially-growing GP stays ahead of the linearly-growing AP forever.

> **Investigate further:** This is Day 1's B10 insight (GP eventually and permanently overtakes AP) turned into a "do they ever meet again" question.

2. **Answer:** A) The binomial row sum grows faster.

Method: Day 6's sum is $\sum_{k=1}^{n} 1 = n$ (linear). Day 2's row sum is 2^n (exponential). Exponential growth eventually and permanently outpaces linear growth.

> **Investigate further:** The exact same "exponential beats linear eventually" theme as C1 and Day 1's B10, now applied to two different sums rather than two sequences.

3. **Answer:** A) Linear growth in n.

Method: A repeated characteristic root of exactly 1 gives closed form $a_n = (A + Bn)(1)^n = A + Bn$ — linear in n (Day 4's C7 result), regardless of how that root of 1 arose.

> **Investigate further:** This shows Day 4's C7 result is really about the VALUE of the repeated root ($= 1$), not about how the recurrence was originally motivated (whether directly given, or arising from a Day 6-style fixed point).

4. **Answer:** A) $\frac{9}{4}$

Method: $(1 - x)^{-2}$ at $x = \frac{1}{3}$: $\left(\frac{2}{3}\right)^{-2} = \left(\frac{3}{2}\right)^{2} = \frac{9}{4}$.

> **Investigate further:** Same technique as B2 and Day 3's C8, now with the "$(k + 1)$-weighted" series instead of the plain geometric series.

5. **Answer:** A) Every starting value returns to itself after exactly 3 applications.

Method: A Möbius transformation of finite order m (as a transformation) satisfies $f^m =$ identity EVERYWHERE it's defined, not just at special points — this is a property of the transformation itself, not of any particular starting value.

> **Investigate further:** Contrast Day 5's C1/C2/D2: those specific maps were shown to have order 4 and 3 respectively by direct computation from ONE starting value, but the underlying fact (guaranteed for every valid starting point) is what this question makes explicit.

6. **Answer:** A) $0 \leq r \leq 1$.

Method: For $0 < r < 1$: decreasing toward 0 (monotonic, convergent). $r = 1$: constant (trivially both). $r = 0$: the sequence is $a, 0, 0, 0, \ldots$, trivially monotonic (non-increasing) and convergent. For $-1 < r < 0$: convergent to 0 but NOT monotonic (alternates sign). For $|r| \geq 1$ (excluding $r = 1$): not convergent (Day 3/B9).

> **Investigate further:** The boundary cases ($r = 0$ and $r = 1$) are exactly where "monotonic" and "convergent" stop being in tension with each other — everywhere else in $(-1, 1)$, oscillation (for negative r) breaks monotonicity even though convergence still holds.

7. **Answer:** A) $|r| < 1, = 1, > 1$ entirely determines the size of r^k.

Method: Both Day 6's search (where $|u| = \frac{c}{r^k-1} \to 0$ because $r \geq 2 > 1$) and Day 4's convergence condition (roots need $|r| < 1$ to shrink to 0) are direct consequences of this single fact about exponentials.

> **Investigate further:** This is the SAME underlying fact used in two superficially different-looking arguments across two different days.

8. **Answer:** A) Geometric series convergence, recurrence convergence, GP boundedness.

Method: All three are direct restatements of "the size of r^n as $n \to \infty$ depends entirely on whether $|r|$ is less than, equal to, or greater than 1" — Day 3's $|r| < 1$ convergence condition, Day 4's root-magnitude convergence condition, and today's B9 boundedness fact are the same underlying idea in three different costumes.

> **Investigate further:** Naming this shared idea explicitly is the whole point of a synthesis day — the individual techniques were taught separately across the week precisely so that noticing they're the same idea, here, lands with real weight.

Section D Challenge

1. **Answer:** 252

Method: Let $d = \gcd(4, n)$. Since the product of every 4 consecutive terms is constant, $a_i = a_{i+4}$ (indices mod n) for every i — so a_i depends only on i mod d. Write $x_0, \ldots, x_{d-1}$ for the values on each residue class. Since $d \mid 4$ always (a gcd divides both its arguments), every window of 4 consecutive terms covers each x_j exactly $4/d$ times, regardless of where the window starts. So the constant product is $(x_0 x_1 \cdots x_{d-1})^{4/d}$, and setting this equal to n: a valid n-necklace exists exactly when n is a perfect $(4/d)$-th power, where $d = \gcd(4, n)$.
Case $4 \mid n$ $(d = 4)$: need n to be a perfect 1st power — always true. Every multiple of 4 works.
Case $n \equiv 2 \pmod 4$ $(d = 2)$: need n to be a perfect square. But every perfect square is $\equiv 0$ or $1 \pmod 4$ ($\text{even}^2 \equiv 0$, $\text{odd}^2 \equiv 1$), never $\equiv 2$ — so this case is NEVER possible.
Case n *odd* $(d = 1)$: need n to be a perfect 4th power (automatically odd, since an odd 4th root gives an odd 4th power).
Counting $4 \leq n \leq 1000$: multiples of 4 number 250. Odd perfect 4th powers in range: $3^4 = 81$ and $5^4 = 625$ (checking $1^4 = 1 < 4$ excluded, $7^4 = 2401 > 1000$ excluded) — 2 more. Total: $250 + 2 = 252$.

> **Investigate further:** The real BMO1 question (2019 Q2) uses windows of 3 instead of 4, over the range $3 \leq n \leq 2018$. Since 3 is prime, $\gcd(3, n) \in \{1, 3\}$ only, giving just two cases (multiples of 3 always work; non-multiples need n to be a perfect cube) — the real answer is 679. Using 4 instead of a prime window introduces the extra $n \equiv 2 \pmod 4$ case, which turns out to NEVER contribute — an example of how changing one number in a BMO1 problem can add real extra casework, not just bigger numbers.

2. **Answer:** Proof by induction using Day 6's construction.

Method: Base case: a_1 given, trivially satisfies $S_1 = a_1$ divisible by 1. Inductive step: given a valid $a_1 < \cdots < a_k$ with S_k divisible by k, Day 6's C7 argument shows some $a_{k+1} \equiv -S_k \pmod{k+1}$ makes S_{k+1} divisible by $k+1$ — and since there are infinitely many integers in that residue class (differing by multiples of $k+1$), we can always choose one that ALSO exceeds a_k (just add enough multiples of $k+1$ to the smallest valid residue). This gives a valid $a_{k+1} > a_k$. By induction, a strictly increasing valid sequence of any desired length n can always be built, starting from any a_1. $\square$

> **Investigate further:** This upgrades Day 6's C7 ("a valid next term always exists") to also guarantee it can be chosen larger than the previous term — the same idea, with one extra degree of freedom (choosing among infinitely many equivalent residues) doing the extra work.

3. **Answer:** A) No — a particular-solution technique is needed instead.

Method: The equation $L = rL - c_n$ would need L to depend on n (since c_n changes every step), contradicting the requirement that L be a single fixed constant — exactly Day 4's C5 situation (a non-constant forcing term defeats the simple constant-shift trick).

> **Investigate further:** Recognising that Day 4's C5 insight (forcing term type determines the fix needed) applies just as much to Day 6's alternating process as to Day 4's own recurrences is real cross-day transfer, not just pattern-completion.

4. **Answer:** $a_1, \ldots, a_5 = 1, 3, 5, 7, 9$; conjectured closed form $a_n = 2n - 1$.

Method: $a_1 = 1$ ($S_1 = 1$). For a_2: try 2 first ($S_2 = 3$, not divisible by 2); try 3 ($S_2 = 4$, divisible by 2) — so $a_2 = 3$. For a_3: try 4 ($S_3 = 8$, not divisible by 3); try 5 ($S_3 = 9$, divisible by 3) — $a_3 = 5$. For a_4: try 6 ($S_4 = 15$, not divisible by 4); try 7 ($S_4 = 16$, divisible by 4) — $a_4 = 7$. For a_5: try 8 ($S_5 = 24$, not divisible by 5); try 9 ($S_5 = 25$, divisible by 5) — $a_5 = 9$.
The pattern $1, 3, 5, 7, 9$ is exactly the consecutive-odd-numbers sequence from Day 6's B1, where $S_k = k^2$ always — and $k^2/k = k$ is always an integer, so this sequence is always valid. The conjecture $a_n = 2n - 1$ is consistent with every computed term, and the fact that the "next even integer" candidate fails at every single step (checked above for $n = 2, 3, 4, 5$) strongly suggests it is genuinely the SMALLEST valid choice at each stage, not a coincidence.

> **Investigate further:** This is Day 6's B1 construction (built there by deliberate choice) rediscovered here as the UNIQUE greedy outcome of a simple local rule ("smallest integer bigger than the last term, keeping the mean an integer") — a satisfying full-circle moment to end the week on.

5. **Answer:** A) Day 2 (Binomial & Trinomial Expansion).

Method: D1's proof rests entirely on periodicity ($a_i = a_{i+4}$, an orbit-structure argument closely related to Day 5's periodicity toolkit) and number-theoretic perfect-power reasoning — binomial coefficients never enter the argument at any point.

> **Investigate further:** Recognising which tools are genuinely load-bearing for a given problem (and which aren't) is itself a mature strategic skill — not every day's toolkit is meant to be forced into every problem, and knowing that saves real time under exam pressure.

Top 5 Patterns — Worksheet #7

1. **Identify which day's toolkit a question wants before diving into algebra.** Misapplying Day 4's heavy machinery to a Day 1/3 problem (or vice versa) wastes real time. *(see A7, B1, B3, B6, B8, C1, C6)*

2. **The single unifying idea behind Days 3, 4, and 5: whether a magnitude is < 1, $= 1$, or > 1 controls everything.** *(see B3, B9, C6, C7, C8)*

3. **Series-manipulation from Day 2 ($(1 - x)^{-m}$ substitutions) combines directly with Day 3's convergence toolkit.** Always look for this bridge on an "evaluate this infinite series" question. *(see B2, C4)*

4. **Day 6's shift-to-fixed-point trick is Day 3's GP toolkit, applied one level up via Day 4's shift technique.** *(see B6, B7, D3)*

5. **Not every technique is relevant to every problem.** Identifying which day's toolkit is genuinely load-bearing (and which isn't) is itself a tested skill on a synthesis day. *(see C2, C3, D5)*

Common Traps — Worksheet #7

1. **Reaching for Day 4's characteristic equation on a recurrence that's secretly just an AP or GP in disguise.** *(see B1, B6)*
2. **Assuming every sum/series question is "the same kind of sum."** Day 2's binomial row sums and Day 6's integer-mean partial sums are structurally unrelated, despite superficial similarity. *(see B10, C2)*
3. **Forgetting that convergence and boundedness are different properties.** $|r| > 1$ makes a GP unbounded, not just "not converging to a specific value." *(see B9)*
4. **Assuming a repeated-root or fixed-point coincidence is special/impossible rather than checking what it actually implies.** *(see C3)*
5. **In a full synthesis proof, skipping the general case-by-case argument in favour of pattern-matching from a few small examples.** A genuine BMO1 proof needs the general argument. *(see D1)*

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."

Found a mistake or want to contribute a sheet?
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